

Enhancing Situational Awareness in Hearing-Impaired Drivers: A Non-Contact Alert System Using Air-Based Tactile Stimulus

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Abstract

Driving presents unique challenges for hearing-impaired individuals, who often miss critical auditory cues and struggle with situational awareness. Most existing assistive systems rely on visual alerts or contact-based tactile feedback. However, visual warnings may distract drivers from the road, and contact-based tactile stimuli lack sufficient spatial resolution and flexibility. In this paper, we propose a non-contact air-based tactile feedback system comprising: (1) a low-cost air-based actuator and (2) a systematic mapping between 16 expressive tactile patterns and driving events tailored to the hearing-impaired context. A driving simulator study with five hearing-impaired participants validated the system's effectiveness: 11 out of 16 stimulus-event mappings achieved over 70% user agreement (i.e., the proportion of participant responses that aligned with the intended mapping exceeded 70%), confirming both perceptual clarity and intuitive design. These results demonstrate the potential of air-based tactile feedback to enhance situational awareness and driving safety for hearing-impaired drivers.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; *Haptic devices*.

Keywords

Non-contact haptics, accessibility, driver safety

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1 Introduction

Driving is a cognitively demanding task [8], and situational awareness (SA) is a critical cognitive capacity for safe driving [5, 9], referring to a driver's ability to perceive environmental elements, comprehend their meaning, and project their future status to inform decision-making. For hearing-impaired individuals, SA is often compromised due to missed auditory cues and challenges in sound localization [3, 9], hindering their driving performance. Hearing-impaired drivers commonly struggle with detecting traffic incidents, identifying vehicle mechanical issues, processing navigational instructions, and responding to phone notifications [6, 15, 18].

Previous research on assisting hearing-impaired drivers has focused on visual warnings [14, 18] and haptic cues [15, 19]. While visual warnings for driving events may divert attention from the road, tactile stimuli maintain visual continuity via an alternative sensory channel [6]. However, many existing tactile alert systems rely on contact-based approaches that offer limited spatial flexibility and feel intrusive. Non-contact methods like air-based haptics offer directional feedback in free space, enabling broader spatial coverage with minimal physical intrusion [6].

To address these limitations, we proposed a non-contact air-based haptic alert system, consisting of an air-based actuator and a systematic mapping between 16 expressive tactile patterns and driving events tailored to the hearing-impaired context. The system uses air-based stimuli to notify hearing-impaired drivers about driving events without interrupting their visual focus. We conducted a two-stage study. First, a co-creation workshop was conducted with six participants who helped select 8 out of 14 designed stimuli for their effectiveness in conveying urgency and spatial proximity. These

were expanded into 16 patterns using single- and bi-directional output from a two-actuator setup. An interview with seven hearing-impaired participants established mappings between each pattern and a relevant driving event. Second, five hearing-impaired drivers assessed the mapping in a validation study.

Results showed that 11 of the 16 event mappings achieved over 70% agreement with the intended categories. These findings highlight the potential of air-based tactile stimuli as effective assistive feedback for hearing-impaired drivers and underscore the importance of comfort and intuitive stimulus-event matching in improving safety.

2 Related Work

Previous research on air-based haptics mostly employs synthetic jet streams generated by speakers and focused through nozzles. The jet stream approach has proven effective as a non-contact tactile interface in various contexts [15, 16]. In the driving domain, visual warnings are commonly used to enhance the situational awareness for hearing-impaired drivers [18], and tactile stimuli are typically delivered via vibration motors in the steering wheel or seat to convey critical information [7].

Shen et al. explored the use of air-jet haptics applied to the driver's head to convey driving-related information [15], demonstrating the feasibility of head-based air stimulation for in-vehicle alerts. Hagerup et al. demonstrated that different tactile waveforms elicit distinct perceptual experiences, highlighting the potential of waveform variation for encoding information [11]. These studies underscore the viability of air-based haptic cues and the importance of stimulus design in communication effectiveness. However, existing work has primarily focused on simple or static air cues, while the use of directional and rhythmic tactile patterns, particularly in the context of hearing-impaired drivers, remains underexplored. To advance this area, we introduce a more expressive set of head-based, directional rhythmic air-based stimuli produced by a cost-effective actuator. Our system is designed to enhance situational awareness without relying on visual or auditory channels.

3 Air-based Haptics

3.1 Actuator Design

We designed a cost-effective, medium-sized haptic actuator suitable for cockpit integration. Based on the synthetic-jet concept from SynJet [15] and the inflated-nozzle approach of AIREAL [16], the actuator uses a 40 W, 4.5 inches speaker paired with a TPA3116 amplifier. For the nozzle, we 3D-printed a series of nozzles varying in diameter (1.5-3.5 cm) and height (0-4.5 cm), see Figure 1.

Tactile evaluations with four participants revealed that larger-diameter nozzles produced stronger jets but blurred spatial resolution, while smaller nozzles offered precise cues at the expense of flow intensity. Nozzle #1 (Ø 3.5 cm × 3.0 cm H) achieved the balance in delivering both sufficient stimulus strength and high directional accuracy, and was therefore selected for the proceeding experiments. The actuator is mounted on an adjustable desk arm to allow fine directional and rotational positioning.

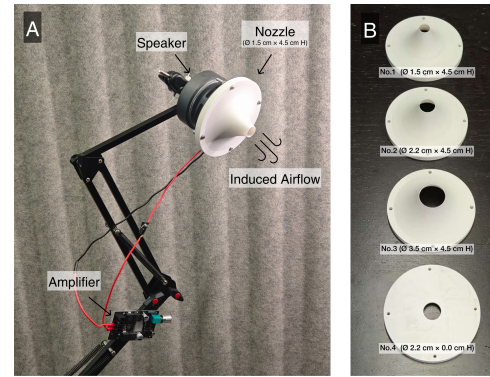


Figure 1: (A) The setup of the actuator. (B) 3D printed nozzles.

3.2 Driving Audio Signal Design

We investigated how air-based tactile rhythms can shape users' perception of urgency and spatial proximity. 14 driving audio signals were designed to produce different air tactile stimuli, based on principles from auditory perception theories, where the increase in sound intensity is known to suggest an approaching source [12], and the tau effect [17] shows that shorter temporal intervals between stimuli are perceived as indicating a closer spatial distance. In automotive sound design, auditory warnings often modulate pitch, loudness, and interval to convey proximity. For example, reversing sensors use accelerating beeps to indicate decreasing distance to nearby obstacles. Based on these parallels, we hypothesized that temporal and rhythmic variations in tactile airflow could convey similar cues related to spatial distance and urgency for hearing-impaired drivers. 14 driving signals (see Table 1, audio files included in the supplementary materials) were constructed using sine waves generated through PyAudio [13] and edited with Adobe Audition [1] (See Figure 2). These signals were composed of rhythmically patterned sine clips, modulated in amplitude and temporal structure to produce distinct perceptual effects.

Table 1: Parameters of the 14 Air-Based Tactile Stimuli

ID	Duration	Gap	Interval	Rhythm Pattern
1	1000 ms	/	1000 ms	Single Pulse
2	1000 ms	/	500 ms	Single Pulse
3	1000 ms	/	100 ms	Single Pulse
4	1000 ms	/	500 ms	Gradient Single Pulse
5	500 ms	/	500 ms	Single Pulse
6	500 ms	/	100 ms	Single Pulse
7	500 ms	/	1000 ms	Single Pulse
8	100 ms	/	300 ms	Single Pulse
9	100 ms	/	100 ms	Single Pulse
10	100 ms	50 ms	500 ms	Triple Pulses
11	100 ms	50 ms	200 ms	Triple Pulse
12	100 ms	/	500 ms	Single Pulse
13	500 ms and 100 ms	50 ms	200 ms	Three-Long-One-Short Pulse
14	10000 ms	/	/	Gradient Blow

3.2.1 Generating sine-wave-based audio. A sine wave with a maximum amplitude of 1 was generated as the original audio. Since most speakers support frequencies above 20 Hz, frequencies from

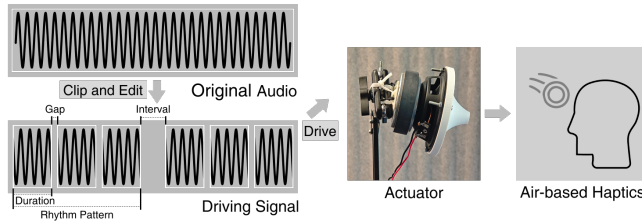


Figure 2: Signal Processing Pipeline for Air-Based Haptics

20 Hz to 60 Hz in 10 Hz increments were tested, with 60 Hz as the upper limit to minimize perceptible noise. Two participants evaluated airflow strength and noise by placing their palms 5 cm above the outlet. Based on tests at 100% computer volume and the amplifier's maximum output, we selected 30 Hz as the optimal driving frequency, offering the strongest and most focused air haptic sensation while maintaining a low and tolerable noise level.

3.2.2 Designing different rhythms. The minimum threshold for perceiving a single tactile stimulus or distinguishing rhythms exceeds 30 ms across all age groups, and that for perceiving gaps between two stimuli also exceeds 30 ms [10]. Given that our air-mediated stimuli are subject to natural dispersion, we set the minimum stimulus duration at 100 ms and the minimum interval at 50 ms to ensure clear perceptual distinction.

4 Co-Creation Workshop

Six participants attended the workshop to evaluate the properties of each tactile stimulus. Each participant was told the general procedure of the experiment, signed a consent form, and sat in a fixed chair with the actuator placed on the right side. The position and rotation of the actuator were adjusted so that the participant felt the tactile stimuli were placed on their right cheek. Each participant was asked to wear noise-canceling headphones to simulate hearing impairment, and to keep their head in the same position to ensure the tactile stimuli stroke the same area of the face. Fig 3 shows the setup of the Co-Creation Workshop.

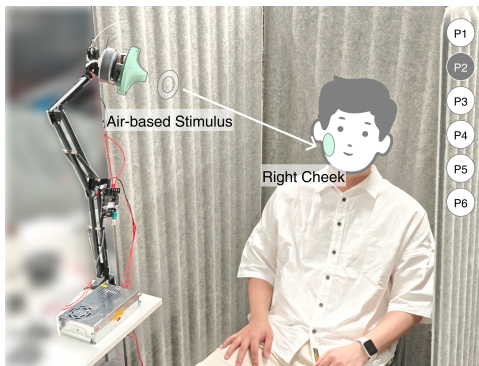


Figure 3: A Participant engaging in the co-creation workshop

After the adjustment was completed, the 14 audio files were played on the actuator. The participant was asked to judge the clarity of the tempo, comfort and urgency on a scale from 1 to 7, meaning from least to most. Their oral feedback on their feelings of the tactile experience of each audio was also recorded.

4.1 Participant Feedback

Shorter intervals or durations, given the same rhythm pattern, increase perceived urgency. Five of six participants rated S_1 – S_3 (1000ms, single pulse, decreasing interval) in increasing urgency; two rated S_3 less urgent than S_2 . Similar trends were observed for S_5 – S_6 (500ms), S_8 – S_9 (100 ms), and S_{10} – S_{11} (triple pulses). P_2 noted that overly fast rhythms felt like continuous breeze, suggesting long durations with short intervals (e.g., S_2 , S_6) may reduce urgency by blurring distinct pulses.

Stronger stimulus lead to higher perceived urgency. Four participants rated S_5 more urgent than S_3 , indicating that stronger stimuli increase perceived urgency.

Longer intervals lead to greater perceived physical distance. For S_7 (500ms, 1000ms interval decreasing by 20%), P_2 – P_6 described a sense of acceleration; P_3 , P_4 , and P_6 compared it to a reversing radar or approaching object. For S_{11} (100ms, 500ms interval decreasing by 20%), all six participants noted increasing tempo, with P_1 , P_3 , P_5 , and P_6 linking it to decreasing distance.

The triple-pulse pattern is perceived as indicating an exceptional or abnormal state. P_1 , P_5 , and P_6 described triple pulses as requiring attention or corrective action. Five participants clearly recognized S_{10} and S_{11} (clarity rating > 5), while only two identified the more complex S_{13} . This suggests triple pulses serve as distinct alerts, whereas complex patterns demand more cognitive effort.

Air-based tactile stimulus suggests spatial direction. In the experiment, the air-based tactile device was consistently placed to the participant's right, directing airflow to their right cheek. P_1 , P_2 , and P_3 associated the sensation with events occurring to their right or front-right, suggesting that the perceived direction of tactile stimulus may implicitly indicate the location of an external event.

5 Tactile Patterns-Driving Events Mapping

5.1 Event Set for Hearing-Impaired Drivers

We constructed a set of traffic-related events relevant to the hearing-impaired driving experience based on semi-structured interviews conducted with seven hearing-impaired drivers who all self-report at least a moderate level of driving experience. The event set includes both common traffic situations and scenarios uniquely encountered by hearing-impaired drivers, and events were further categorized into four types based on their characteristics. We categorize the events into four main categories, see Table 2.

Findings from the interviews are as follows. First, drivers with hearing impairments rely primarily on visual input, making it difficult to perceive events outside their field of view. Second, many are unable to localize sound sources even with hearing devices.

Relational Mapping Hypotheses

Based on insights from the co-creation workshop, we selected eight air-based tactile stimuli (S_2 – S_9 is S_2 played at -10dB) and designed each to be presented in a single-directional or bi-directional mode, resulting in 16 tactile variations. We believe these stimuli represent different types of events from the constructed event set. Their correspondence to specific event categories is shown in Table 4.

5.2 Validation Study

To validate the mapping between the stimuli patterns and events categories, we invited 5 hearing-impaired drivers who have least

Table 2: Categorized Driving Event Set

Immediate Traffic Hazards	Vehicle Condition Alerts
Pedestrian Crossing	Engine Failure
Sudden Cut Lane	Out of Oil / Battery
Crashes	Brakes Failure
Emergency Vehicle	
Interaction Notifications	Navigation and Guidance
Backseat Passenger Interaction	Rear Parking Sensors
Unfastened Seatbelt	Highway Exit
Engine Start	Traffic Light Turning
Scenic Viewpoint Ahead	Construction Ahead

two years of driving experience and have been driving regularly for the past six months to a matching workshop. Detailed participant information is provided in Table 3. The classification of hearing loss degree follows WHO standards [4]. The workshop began with the participant signing consent forms and being told the procedure of the experiment. The participant was given images with captions about different driving events relevant to the context, listed in Table 2. The participant was told they could suggest additional events.

Table 3: Hearing Impaired Participant Profiles

Participant Driver	Hearing Loss Degree	Hearing Device	Driving Experience
D_1	Mild	Hearing Aid	6 years
D_2	Moderate	Hearing Aid	8 years
D_3	Severe	Hearing Aid	5 years
D_4	Severe	Cochlear Implant	2 years
D_5	Severe	Cochlear Implant	7 years

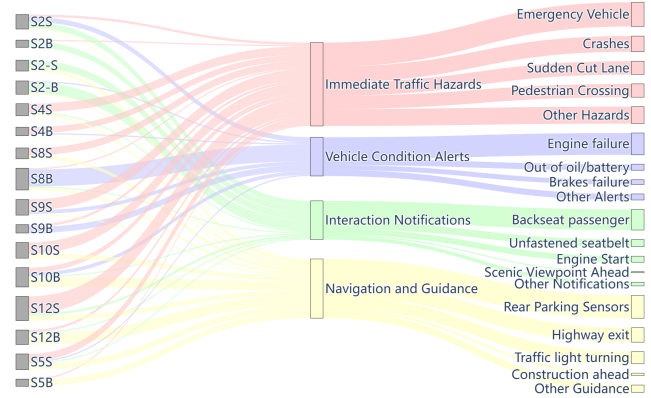
Participants sat in a SCANer driving simulator [2]. After 5 minutes of driving, the simulation paused with the last frame displayed. Two actuators mounted beside the driver's seat were adjusted until cheek haptic perception was reported. Participants then correlated tactile stimuli with three driving events, referencing event images with multiple stimulus exposures allowed.

5.3 Participant Feedback

Participants achieved >70% agreement on 11/16 initial tactile pattern-to-event category mappings. The agreement rate for a specific mapping was defined as the proportion of trials in which participants selected the intended event category for that particular tactile pattern. Three mappings showed no majority agreement (<50%), while two failed to match the initial mapping (Table 4).

6 Conclusion and Future Work

This research introduced a non-contact air-based tactile alert system designed to enhance the situational awareness of hearing-impaired drivers. Through a co-design process with general participants to explore basic haptic rhythms, we found that eight out of the 14 rhythmic air-based stimuli effectively conveyed urgency and


Figure 4: Validation Results in Sankey Diagram
Table 4: Initial Mapping and Participant Agreement of Tactile Patterns - Driving Events

Item	Single Direction		Both Direction	
	Initial	Participant	Initial	Participant
S_2	N&G	No Majority	IN	IN
S_{2-}	IN	IN	IN	IN
S_4	ITH	ITH	ITH	ITH
S_8	N&G	ITH	VCA	VCA
S_9	ITH	ITH	VCA	VCA
S_{10}	N&G	N&G	VCA	No Majority
S_{12}	N&G	ITH	N&G	N&G
S_5	N&G	No Majority	N&G	N&G

N&G: Navigation and Guidance; IN: Interaction Notifications; VCA: Vehicle Condition Alerts; ITH: Immediate Traffic Hazards

spatial proximity. The eight stimuli were developed into 16 tactile patterns by varying single-directional or bi-directional output in a two-actuator setup. Interviews with hearing-impaired drivers were conducted to establish a mapping between four main driving event categories to each of the 16 tactile patterns. In the validation study, 11 out of 16 proposed stimulus–event pairings achieved over 70 percent agreement with initial driving event categories. These findings show the potential of an air-based feedback system with a tactile pattern-driving event mapping as an effective assistive alert system for hearing-impaired drivers.

Despite promising results, our current work remains at the prototype stage. Evaluations in this study were conducted in static settings; real-world driving introduces additional challenges such as cabin airflow and ambient noise. Future work will address these limitations through both high-fidelity driving simulations and on-road trials using the Wizard of Oz method to emulate fully integrated system behavior. Further iterations will refine stimulus parameters, expand the event taxonomy, and explore user-customizable mappings and audio parameters.

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